

MA443 Homework 3

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Problem 1

Statement

Maximize $f(\bar{x}) = x_1 + 2x_2$ subject to the following inequality constraints:

$$3x_1 + 4x_2 \leq 12$$

$$x_1 + 3x_2 \geq 3$$

$$x_1 \geq 0$$

- Solve this problem analytically using the KKT conditions and ensure the necessary conditions are satisfied.
- Plot the inequality constraints and the contours of L in MATLAB and shade (by hand) the feasible region. Solve the problem using "linprog". Include the optimal point (x_1, x_2) , the optimal cost, and the Lagrange multipliers.

Analytical Solution

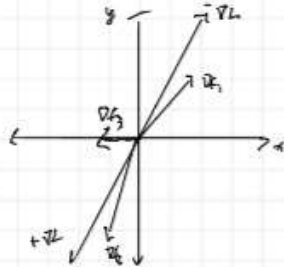
$$\max f(\bar{x}) = x_1 + 2x_2 \text{ subject to } \bar{f}$$

$$\bar{f}_1: 3x_1 + 4x_2 \leq 12 \rightarrow 3x_1 + 4x_2 - 12 \leq 0$$

$$\bar{f}_2: x_1 + 3x_2 \geq 3 \rightarrow -x_1 - 3x_2 + 3 \leq 0$$

$$\bar{f}_3: x_1 \geq 0 \rightarrow -x_1 \leq 0$$

Linear problem, to determine active constraints, simplex method is used:



$$L(\bar{\mu}) = -(x_1 + 2x_2) \quad \bar{x} = [x_1, x_2]$$

$$\frac{\partial L}{\partial \bar{x}} = [-1 \ -2]$$

$$\frac{\partial \bar{f}}{\partial \bar{x}} = \begin{bmatrix} 3 & 4 \\ -1 & -3 \\ -1 & 0 \end{bmatrix}$$

$\bar{f}_1, \bar{f}_2$ must be active $\rightarrow \bar{f}_1, \bar{f}_2$ is active constraints.

$$H = L + \lambda \bar{f} = -x_1 - 2x_2 + \lambda_1 (3x_1 + 4x_2 - 12) + \lambda_2 (-x_1)$$

$$\frac{\partial H}{\partial \bar{x}} = \bar{0} = \begin{bmatrix} -1 + 3\lambda_1 - \lambda_2 \\ -2 + 4\lambda_1 \end{bmatrix} \quad \left. \begin{array}{l} 4 \text{ lines} \\ 4 \text{ unknowns} \end{array} \right\}$$

$$\frac{\partial H}{\partial \lambda} = \bar{0} = \begin{bmatrix} 3x_1 + 4x_2 - 12 \\ -x_1 \end{bmatrix}$$

$$\begin{bmatrix} x_1^* \\ x_2^* \\ \lambda_1^* \\ \lambda_2^* \end{bmatrix} = \begin{bmatrix} 0 & 0 & 3 & -1 \\ 0 & 0 & 4 & -3 \\ 3 & 4 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 2 \\ 12 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 1/2 \\ 1/2 \end{bmatrix}$$

$$\lambda_2 = 0 \rightarrow \bar{\lambda} = \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \end{bmatrix} \geq 0 \text{ is KKT condition satisfied.}$$

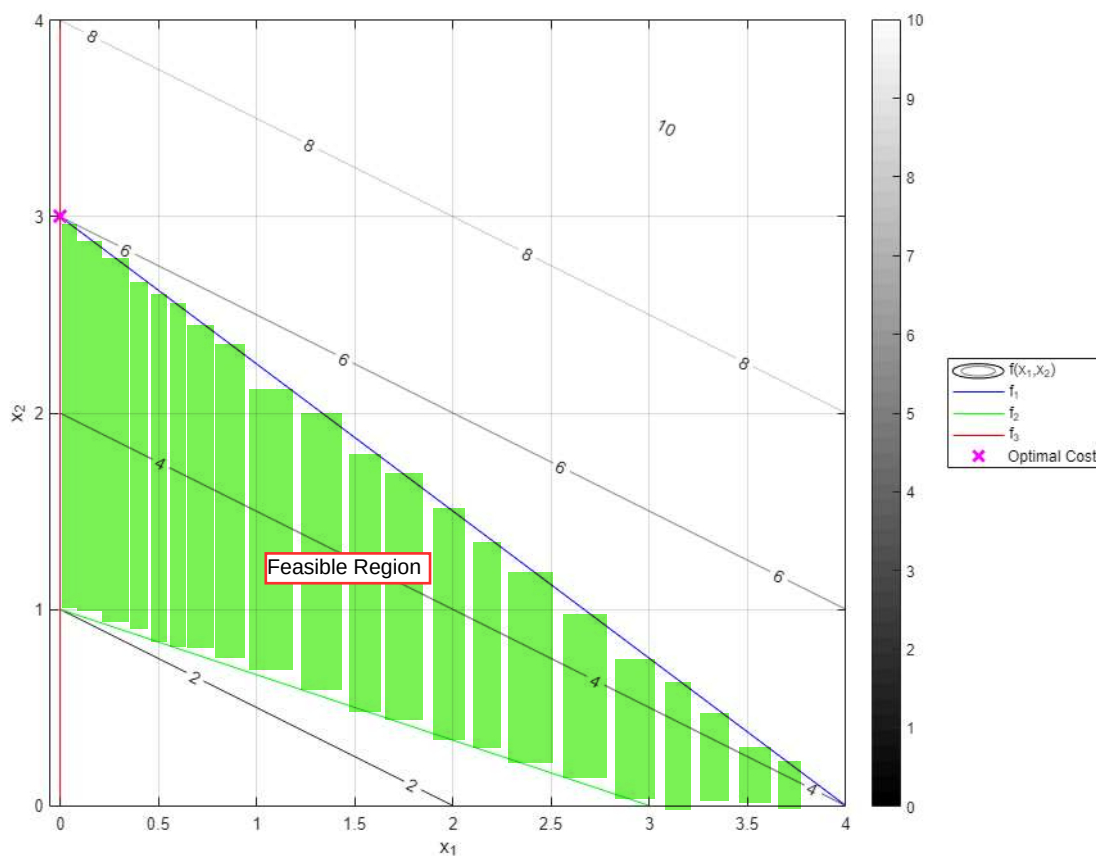
$$L^*(\bar{\mu}) = -(x_1 + 2x_2) = -6 \rightarrow \max f(\bar{x}) = 6.$$

MATLAB Solution

```

clc; clf;
[x1,x2] = meshgrid(0:0.1:10,0:0.1:10);
f = x1+2*x2;
[C,h] = contour(x1,x2,f, [0:2:10], 'ShowText','on');
colorbar
colormap gray
axis equal
axis manual
hold on
grid on
plot([0,4],[3,0], '-b', [0,3],[1,0], '-g', [0,0],[0,10], '-r')
plot(0,3,'mx','MarkerSize', 10, 'LineWidth', 2)
xlim([-0.05,4])
yticks([0:4])
ylim([0,4])
xlabel('x_1')
ylabel('x_2')
%fill([0,0,4,3],[1,3,0,0], 'g', 'FaceAlpha', 0.5)
legend('f(x_1,x_2)', 'f_1', 'f_2', 'f_3', 'Optimal Cost', 'Location', 'eastoutside')
set(gcf, 'Position', [100, 100, 1000, 800]);

```



Solution using linprog(), optimal value is $X = [0, 3]$, optimal cost is -6 (inverse for maximization is 6); both agree with analytical solution.

```
[X,FVAL,EXITFLAG,OUTPUT,LAMBDA] = linprog([-1;-2],[3,4;-1,-3;-1,0],[12;-3;0])
```

Optimal solution found.

X = 2x1

0

3

FVAL = -6

EXITFLAG = 1

OUTPUT = struct with fields:

iterations: 2

constrviolation: 0

message: 'Optimal solution found.'

algorithm: 'dual-simplex'

firstorderopt: 1.1102e-16

```
LAMBDA = struct with fields:
    lower: [2x1 double]
    upper: [2x1 double]
    eqlin: []
    ineqlin: [3x1 double]
```

Lagrange Multipliers:

```
LAMBDA.ineqlin
```

```
ans = 3x1
    0.5000
         0
    0.5000
```

Problem 2

Statement

Minimize:

$$f(x) = (x_1 - 4)^2 + (x_2 - 8)^2$$

subject to:

$$x_1 + x_2 \leq 12$$

$$x_2 \leq 5$$

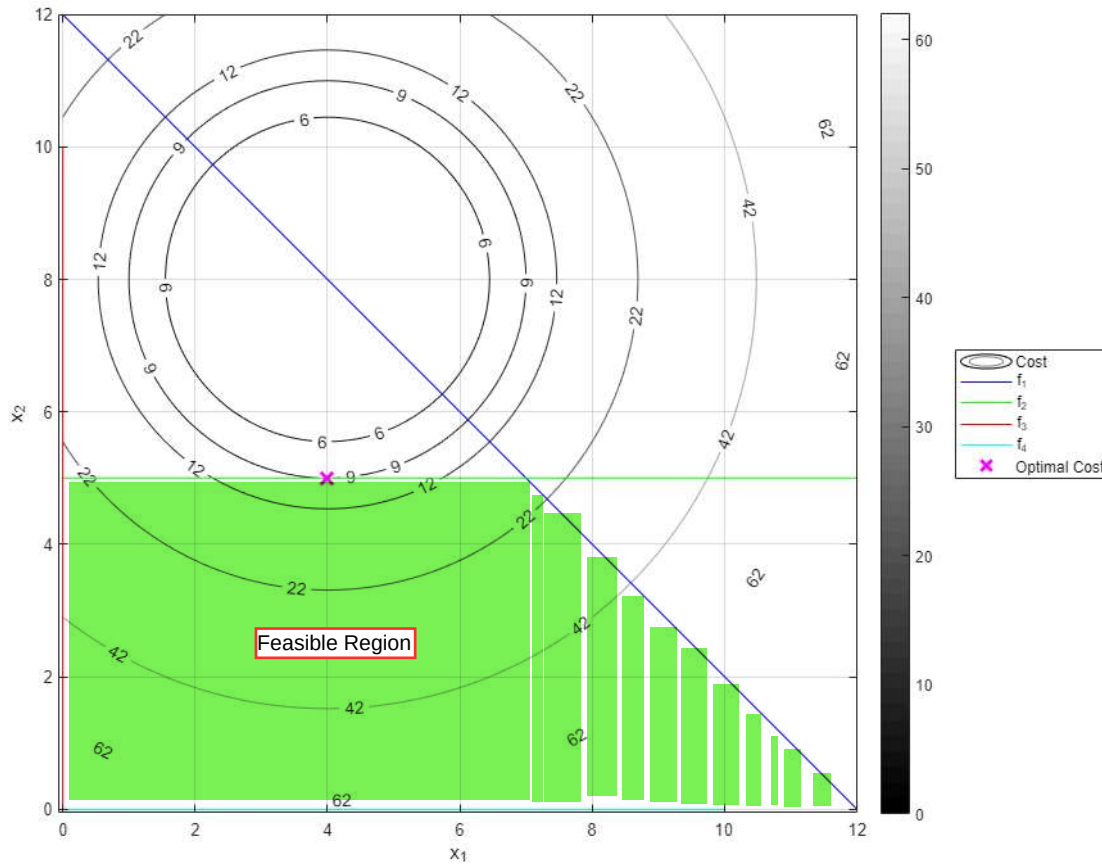
$$x_1 \geq 0$$

$$x_2 \geq 0$$

- Plot the inequality constraints in MATLAB and shade (by hand) the feasible region. Also plot (in the same figure) the contours of $L(\bar{y})$. Solve the problem using “fmincon”. Include the optimal point (x_1, x_2) , the optimal cost, and the Lagrange multipliers.
- Solve this problem analytically using the KKT conditions and ensure the necessary conditions are satisfied.

MATLAB Solution

```
clc; clf;
[x1,x2] = meshgrid(0:0.1:12,0:0.1:12);
L = (x1-4).^2+(x2-8).^2;
[C,h] = contour(x1,x2,L, [0:6:12, 22:20:62, 9], 'ShowText','on');
colorbar
colormap gray
axis equal
axis manual
hold on
grid on
plot([0,12],[12,0], '-b',[0,15],[5,5], '-g',[0,0],[0,10], '-r',[0,10],[0,0], '-c')
plot(4,5,'mx','MarkerSize', 10, 'LineWidth', 2)
xlim([-0.05,12])
yticks([0:2:12])
ylim([-0.05,12])
xlabel('x_1')
ylabel('x_2')
%fill([0,0,4,3],[1,3,0,0], 'g','FaceAlpha',0.5)
legend('Cost','f_1','f_2','f_3','f_4','Optimal Cost','Location','eastoutside')
```



Solution via `fmincon()`, optimal point is `[4 5]` and associated optimal cost is 9.

```
[X1,FVAL1,EXITFLAG,OUTPUT,LAMBDA,GRAD,HESSIAN] = fmincon(@(p)(p(1)-4)^2+(p(2)-8)^2,[4,4],[1,1;0,1;-1,0;0,-1],[12;5;0;0])
```

Local minimum found that satisfies the constraints.

Optimization completed because the objective function is non-decreasing in feasible directions, to within the value of the optimality tolerance, and constraints are satisfied to within the value of the constraint tolerance.

<stopping criteria details>

X1 = 1×2

4.0000 5.0000

FVAL1 = 9.0000

EXITFLAG = 1

OUTPUT = struct with fields:

iterations: 4

funcCount: 15

constrviolation: 0

stepsize: 1.6720e-04

algorithm: 'interior-point'

firstorderopt: 3.3422e-06

cgiterations: 0

message: 'Local minimum found that satisfies the constraints.' Optimization completed because the objective function is non-decreasing in feasible d.

bestfeasible: [1×1 struct]

LAMBDA = struct with fields:

eqlin: [0×1 double]

eqnonlin: [0×1 double]

ineqlin: [4×1 double]

lower: [2×1 double]

upper: [2×1 double]

ineqnonlin: [0×1 double]

GRAD = 2×1

-0.0000

-6.0000

HESSIAN = 2×2

2.0000 -0.0001

-0.0001 2.0002

Lagrange Multipliers

LAMBDA.ineqlin

```
ans = 4x1
    0.0000
    6.0000
    0.0000
    0.0000
```

Analytical Solution

Graphically, 1 constraint active

$$f_2 = x_2 \leq 5 \rightarrow x_2 - 5 \leq 0$$

$$Z(\bar{x}) = (x_1 - 4)^2 + (x_2 - 8)^2$$

$$\bar{x} = x_1, \quad \bar{u} = x_2$$

$$H = L + \bar{\lambda} \bar{f} = (x_1 - 4)^2 + (x_2 - 8)^2 + \lambda_2 (x_2 - 5)$$

$$\begin{aligned} \frac{\partial H}{\partial x_1} &= \frac{\partial H}{\partial x_1} = 2(x_1 - 4) = 0 \\ \frac{\partial H}{\partial u} &= \frac{\partial H}{\partial x_2} = 2(x_2 - 8) + \lambda_2 = 0 \\ \frac{\partial H}{\partial \lambda} &= \frac{\partial H}{\partial \lambda_2} = x_2 - 5 = 0 \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} \text{3 eqns (linear)} \\ \text{3 vars} \end{array}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 8 \\ 16 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$\begin{aligned} \lambda_1 &= \lambda_3 = \lambda_4 = 0; \text{ all inactive} \\ \lambda_2 &= 6 \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \bar{\lambda} \geq 0 \therefore \text{1st satisfied.}$$

Problem 3

MATLAB Solution

```
clc; clf;
```

Solution via fmincon(), optimal point is [2.9499 1.2247] and associated optimal cost is 36.1783.

```
[X2,FVAL2,EXITFLAG,OUTPUT,LAMBDA,GRAD,HESSIAN] = fmincon(@(p) 3*p(1)^2-2*p(1)*p(2)+5*p(2)^2+8*p(2),[1.73737,0],[],[],[],[],[],[],@nonlcon)
```

Local minimum found that satisfies the constraints.

Optimization completed because the objective function is non-decreasing in feasible directions, to within the value of the optimality tolerance, and constraints are satisfied to within the value of the constraint tolerance.

<stopping criteria details>

X2 = 1x2

2.9499 1.2247

FVAL2 = 36.1783

EXITFLAG = 1

OUTPUT = struct with fields:

iterations: 7

funcCount: 27

constrviolation: 0

stepsize: 1.1610e-05

algorithm: 'interior-point'

firstorderopt: 7.7396e-07

cgiterations: 0

message: 'Local minimum found that satisfies the constraints.' "Optimization completed because the objective function is non-decreasing in "feasible d.

bestfeasible: [1x1 struct]

LAMBDA = struct with fields:

eqlin: [0x1 double]

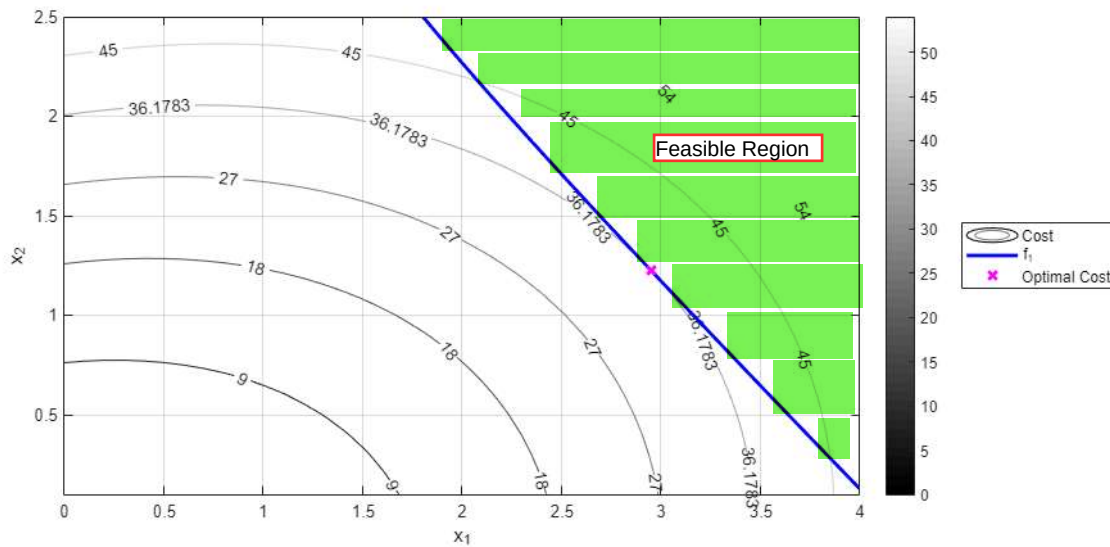
eqnonlin: [0x1 double]

```

ineqlin: [0x1 double]
lower: [2x1 double]
upper: [2x1 double]
ineqnonlin: 2.5848
GRAD = 2x1
    15.2502
    14.3473
HESSIAN = 2x2
    0.8464    -2.0153
   -2.0153    15.0874

[x1,x2] = meshgrid(0:0.1:50,0:0.1:50);
L = 3*x1.^2 - 2*x1.*x2 + 5*x2.^2 + 8*x2;
[C,h] = contour(x1,x2,L, [0:9:27,45:9:54, FVAL2], 'ShowText', 'on');
colorbar
colormap gray
axis equal
axis manual
hold on
grid on
%fplot(@(p)-(p(1)^2-p(2)^2+8*p(2))+17)
ycon = linspace(-10, 10, 1000);
x_right = sqrt(ycon.^2 - 8*ycon + 17);
plot(x_right, ycon, 'b', 'LineWidth', 2);
plot(X2(1),X2(2),'mx','MarkerSize', 7, 'LineWidth', 2)
xlim([0,4])
ylim([0.1,2.5])
xlabel('x_1')
ylabel('x_2')
%fill([0,0,4,3],[1,3,0,0], 'g', 'FaceAlpha',0.5)
legend('Cost','f_1','Optimal Cost','Location','eastoutside')

```



```
set(gcf, 'Position', [100, 100, 1000, 800]);
```

Analytical Solution

$$\mathcal{L}(\bar{p}) = 3x_1^2 - 2x_1x_2 + 5x_2^2 + 8x_2 \quad \bar{x} = x_1, \bar{u} = x_2$$

$$\bar{f} = f_1 = x_1^2 - x_2^2 + 8x_2 \geq 17$$

$$H = \mathcal{L} + \lambda \bar{f} = 3x_1^2 - 2x_1x_2 + 5x_2^2 + 8x_2 + \lambda_1 (17 - x_1^2 + x_2^2 - 8x_2)$$

$$\left. \begin{aligned} \frac{\partial H}{\partial x_1} = \frac{\partial H}{\partial x_1} &= 6x_1 - 2x_2 - 2\lambda_1 x_1 \\ \frac{\partial H}{\partial x_2} = \frac{\partial H}{\partial x_2} &= -2x_1 + 10x_2 + 2\lambda_1 x_2 + 8 - 8\lambda_1 \\ \frac{\partial H}{\partial \lambda} = \frac{\partial H}{\partial \lambda} &= 17 - x_1^2 + x_2^2 - 8x_2 \end{aligned} \right\} \begin{array}{l} 3 \text{ nonlinear eqns} \\ 3 \text{ v.ks.} \end{array}$$

Solve in MATLAB

$$\begin{bmatrix} x_1^* \\ x_2^* \\ \lambda_1^* \end{bmatrix} = \begin{bmatrix} 2.9499 \\ 1.2247 \\ 2.5848 \end{bmatrix} \quad \bar{\lambda} \geq 0 \therefore \text{KKT satisfied.}$$

$$\frac{\partial^2 \mathcal{L}}{\partial u^2} = L_{uu} = \begin{bmatrix} z^* & I_m \end{bmatrix} \begin{bmatrix} H_{xx} & H_{xu} \\ H_{ux} & H_{uu} \end{bmatrix} \begin{bmatrix} z \\ I_m \end{bmatrix}$$

$$\text{where } z = -(t_x)^{-1} f_u$$

$$z = -(2x_1, 1)'' \cdot (-2x_2, 8)$$

$$z = -0.440812$$

$$L_{uu} = \begin{bmatrix} -0.4408 & 1 \end{bmatrix} \begin{bmatrix} 6-2\lambda_1^* & -2 \\ -2 & 10+2\lambda_1^* \end{bmatrix} \begin{bmatrix} -0.440812 \\ 1 \end{bmatrix} = 19.6678$$

$$\text{Let } \lambda_1^* = 2.5848$$

$\geq 0 \checkmark$
 $\therefore \text{is a min}$

```
pAnalytical = fsolve(@(p)[6*p(1)-2*p(2)-2*p(1)*p(3); -2*p(1)+10*p(2)+2*p(2)*p(3)+8-8*p(3); 17-p(1)^2+p(2)^2-8*p(2)], [3, 1.5, 0])
```

Equation solved.

fsolve completed because the vector of function values is near zero as measured by the value of the function tolerance, and the problem appears regular as measured by the gradient.

<stopping criteria details>

pAnalytical = 1x3

2.9499 1.2247 2.5848

```
function [c,ceq] = nonlcon(p)
c = 17-(p(1)^2-p(2)^2+8*p(2)); %<= form
ceq = [];
end
```